SK hynix Debuts First GDDR6-AiM Accelerator Card 'AiMX'

Source URL: https://news.skhynix.com/sk-hynix-debuts-first-gddr6-aim-accelerator-card-aimx-for-generative-ai/

Business

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SK hynix Debuts Prototype of First GDDR6-AiM Accelerator Card 'AiMX' for Generative AI

September 18, 2023

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SK hynix unveiled and demonstrated a prototype of AiMX

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, a generative AI accelerator

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card based on GDDR6-AiM, at the AI Hardware & Edge AI Summit 2023 held September 12–14 at the Santa Clara Marriott, California.

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Accelerator-in-Memory based Accelerator (AiMX)

: SK hynix's accelerator card product that specializes in large language models (AI that learns with large amounts of text data such as ChatGPT) using GDDR6-AiM chips.

2

Accelerator

: A special-purpose hardware device that uses a chip designed specifically for processing and computing information.

Figure 1. SK hynix's exhibition booth at the AI Hardware & Edge AI Summit 2023

Hosted annually by the UK marketing firm Kisaco Research, the AI Hardware & Edge AI Summit brings together global IT companies and high-profile startups to share their developments in artificial intelligence and machine learning. This is SK hynix's third time participating in the summit.

At the event, the company showcased the prototype of AiMX, an accelerator card that combines multiple GDDR6-AiMs to further enhance performance, along with the GDDR6-AIM itself under the slogan of "Boost Your AI: Discover the Power of PIM

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with SK hynix's AiM

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Processing-In-Memory (PIM)

: A next-generation technology that adds computational capabilities to semiconductor memories to solve the problem of data movement congestion in AI and big data processing.

4

Accelerator in Memory (AiM)

: SK hynix's PIM semiconductor product name, which includes GDDR6-AiM.

Figure 2. The prototype AiMX card utilizes multiple GDDR6-AiM chips for enhanced performance

As a low-power, high-speed memory solution capable of handling large amounts of data, AiMX is set to play a key role in the advancement of data-intensive generative AI

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systems. The performance of generative AI improves as it is trained on more data, highlighting the need for high-performance products which can be applied to an array of generative AI systems.

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Generative AI

: AI that learns from large amounts of data to actively generate results based on a user's specific needs.

Figure 3. Demonstrating a large AI language model with AiMX that utilizes GDDR6-AiM

SK hynix also demonstrated Meta's generative AI Open Pretrained Transformer (OPT) 13B model on a server system equipped with the AiMX prototype. The AiMX system featuring GDDR6-AiM chips reduces data processing time by more than 10 times compared to systems with GPUs, while consuming one-fifth the power. The company's demonstration piqued the interest of global companies providing AI services by showing that it can deliver higher performance
6
than the most recent accelerators.
6
Performance is based on the condition that the AiM Control Hub inside the AiMX card is developed as an application-specific integrated circuit (ASIC).
Figure 4. Eui-cheol Lim, vice president of SK hynix's Solution Development division, delivers a presentation on AiMX
In addition, the company held a session outlining the benefits of AiMX. In a presentation titled "Cost-Effective Generative AI Inference Acceleration using AiM," Eui-cheol Lim, vice president of the Solution Development division, compared the performance of GPUs and AiMX and discussed the future of next-generation intelligent semiconductor memories.
"SK hynix's AiMX is a solution that delivers higher performance while consuming less power, and costing less than conventional GPUs," Lim explained. "We will continue to develop memory technologies that will lead the way in the era of artificial intelligence."
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